

Solution Problem 8.2 (see also m-file)

$$\text{PCA}(\Sigma): \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix}$$

$$\lambda: \quad |\Sigma - \lambda I| = 0$$

$$\Leftrightarrow (\sigma_1^2 - \lambda)(\sigma_2^2 - \lambda) - \rho^2 \sigma_1^2 \sigma_2^2 = 0$$

$$\Leftrightarrow \lambda^2 - (\sigma_1^2 + \sigma_2^2)\lambda + (1 - \rho^2)\sigma_1^2 \sigma_2^2 = 0$$

$$\Leftrightarrow \lambda = \frac{1}{2}(\sigma_1^2 + \sigma_2^2) \pm \frac{1}{2}\sqrt{(\sigma_1^2 - \sigma_2^2)^2 + (2\rho \sigma_1 \sigma_2)^2}$$

$$e: \quad \Sigma e = \lambda e$$

$$\Leftrightarrow \begin{cases} (\sigma_1^2 - \lambda)e_1 + \rho \sigma_1 \sigma_2 e_2 = 0 \\ \rho \sigma_1 \sigma_2 e_1 + (\sigma_2^2 - \lambda)e_2 = 0 \end{cases}$$

$$\Leftrightarrow e_2 = \frac{\lambda - \sigma_1^2}{\rho \sigma_1 \sigma_2} e_1$$

$$\Leftrightarrow e = \begin{bmatrix} 1 \\ \frac{\lambda - \sigma_1^2}{\rho \sigma_1 \sigma_2} \end{bmatrix} / \sqrt{1 + \left(\frac{\lambda - \sigma_1^2}{\rho \sigma_1 \sigma_2}\right)^2}$$

$$\text{Var explained} = \frac{\lambda}{\sigma_1^2 + \sigma_2^2}$$

$$\text{PCA}(\rho) \quad \rho = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$

$$\lambda: \quad |\rho - \lambda I| = 0$$

$$\lambda = 1 \pm \rho$$

$$e \quad \rho > 0 \Rightarrow e = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\rho < 0 \Rightarrow e = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\text{Var explained} = \lambda/2$$

Solution Problem 8.3 (see also m-file)

1a) EVD(S): $S = \sum_{i=1}^p \lambda_{s,i} e_i e_i^T = E_S \Lambda_S E_S^T = \begin{bmatrix} \uparrow \\ e_i \\ \downarrow \end{bmatrix}_{p \times p} \begin{bmatrix} \nwarrow \lambda_{s,i} \nearrow \\ \end{bmatrix}_{p \times p} \begin{bmatrix} \leftarrow \\ e_i \\ \rightarrow \end{bmatrix}_{p \times p}$

$(\lambda_{s,i}, e_i) = \text{eig}(S)$

$\Downarrow$
 $\text{PC}_i \text{ loadings} = e_i$
 $\text{PC}_i \text{ scores} = X \cdot e_i$
 $\text{PC}_i \text{ variance} = \lambda_{s,i}$

1b) SVD(Y): $Y \stackrel{\text{def}}{=} \frac{1}{\sqrt{n-1}} X = \sum_{i=1}^p \lambda_{Y,i} u_i v_i^T = U D P^T = \begin{bmatrix} \uparrow \\ u_i \\ \downarrow \end{bmatrix}_{n \times p} \begin{bmatrix} \nwarrow \lambda_{Y,i} \nearrow \\ \end{bmatrix}_{p \times p} \begin{bmatrix} \leftarrow \\ v_i \\ \rightarrow \end{bmatrix}_{p \times p}$

Since $Y^T Y = \frac{1}{n-1} X^T X = S$

and $Y^T Y = P D^T U^T U D P^T = P D^2 P^T$

$\Downarrow$ $\text{eig}(Y^T Y) = \text{eig}(S) = (\lambda_{Y,i}^2, v_i)$

$\Downarrow$
 $\text{PC}_i \text{ loadings} = v_i = e_i$
 $\text{PC}_i \text{ scores} = X v_i = X e_i$
 $\text{PC}_i \text{ variance} = \lambda_{Y,i}^2$

QED

2a) EVD(R): $R = \sum_{i=1}^p \lambda_{R,i} e_i e_i^T = E_R \Lambda_R E_R^T$

$(\lambda_{R,i}, e_i) = \text{eig}(R)$

$\Downarrow$
 $\text{PC}_i \text{ Z-loadings} = e_i$
 $\text{PC}_i \text{ X-loadings} = \sqrt{1/2} e_i$
 $\text{PC}_i \text{ scores} = X \sqrt{1/2} e_i$
 $\text{PC}_i \text{ variance} = \lambda_{R,i}$

2b) SVD(Z): $Z \stackrel{\text{def}}{=} \frac{1}{\sqrt{n-1}} X V^{-1/2} = \sum \lambda_{Z,i} u_i v_i^T = U D P^T$

Since $Z^T Z = \frac{1}{n-1} V^{-1/2} X^T X V^{-1/2} = R$

$\Downarrow$ and $Z^T Z = P D^2 P^T$

$\text{eig}(Z^T Z) = \text{eig}(R) = (\lambda_{Z,i}^2, v_i)$

$\text{PC}_i \text{ Z-loadings} = v_i = e_i$
 $\text{PC}_i \text{ X-loadings} = \sqrt{1/2} e_i$
 $\text{PC}_i \text{ scores} = X \sqrt{1/2} e_i$
 $\text{PC}_i \text{ variance} = \lambda_{Z,i}^2$

QED